

A Most Favored Nation: Augmented Reality and Immersive Performance

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Figure 1: Six unretouched audience views of the same moment, seen through the *Grasshopper* AR application. (AR elements are the most colorful, while the in-person performance seen via the pass-through camera is desaturated.)

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1 INTRODUCTION

A Most Favored Nation was an immersive theatrical performance with augmented reality (AR) set in the world of the Amazon Studios series *The Man in the High Castle*. In the performance, AR was incorporated directly into the original story as a view into another world that inspires action in the characters. Like the series—and the book upon which it is based, by Philip K. Dick—the in-person performance was set in a world where the Axis powers have won World War II. In the novel, characters learn about an alternate universe,

similar to our own, through a book called *The Grasshopper Lies Heavy*. In the streaming series, the book becomes a set of newsreels showing various alternative timelines. In *A Most Favored Nation*, the *Grasshopper* is a personal television that has been “hacked” to show another world existing in parallel with our own. Such devices are carried by the audience and some characters. Figure 1 shows examples of various live views of a single moment in the performance from different audience *Grasshoppers*.

In our production, the audience circulated with their *Grasshoppers* as commoners, moving among three storylines playing out simultaneously. Each storyline featured two parallel versions of 1963 Chicago that overlapped in space and time via augmented reality. The “Axis” timeline was on the physical stage, and the “Allied” version was in AR. In total, the team created twenty-eight original scenes, in which characters grapple with both their immediate circumstances and the knowledge of another life for themselves and those they love. A typical audience experience was about an hour. Due to COVID-19 restrictions, the piece was performed “live-to-tape” with small audiences, who also recorded it for later assembly into an interactive streaming film.

Challenges that we encountered in creating the performance included: 1) Balancing the audience agency encouraged by AR with

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Figure 2: Audience view of a character on-stage looking at an alternative version of himself through the *Grasshopper*.



Figure 3: Operator view showing fiducials, video, physical set previsualization, AR set, and live AR device position.

a typical spectator experience; 2) Coordinating parallel experience of digital and physical worlds, each with actors' performances; 3) Achieving a look-and-feel that felt familiar to viewers of the series; 4) Integrating theater systems for lighting, sound, and communications; 5) Managing the large, multidisciplinary creative process during the COVID-19 pandemic; and, as a result of the pandemic, 6) Capturing and representing the live experience faithfully to audiences that couldn't attend in person. Our approach to these challenges will be discussed in the demonstration, including how they evolved from our previous experience in an earlier AR theater workshop project, *Apprentice at the End of the World*¹.

2 XR DEMONSTRATION

Our demonstration will share sample AR scenes seen live through *Grasshoppers*, with an opportunity for discussion on the story, process, technology, challenges, and future plans. Each cycle will last about fifteen minutes, including time for discussion. Attendees can use their own device or we will provide one. The mix of physical and AR elements will include the following:

Grasshopper AR app. Using a streamlined version of the multiplayer AR application, built in Unreal Engine for iPhone 11 Pro

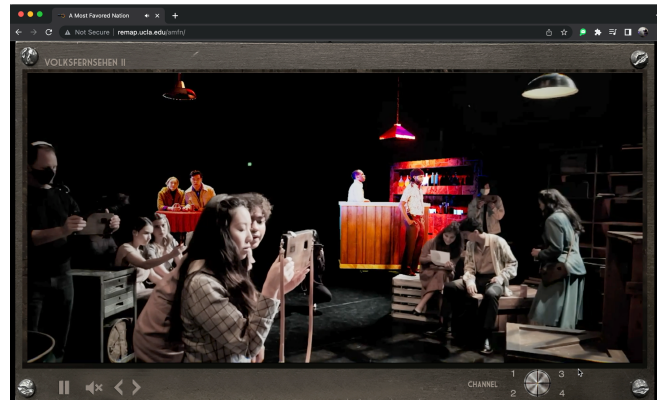


Figure 4: An interactive "live-to-tape" film was shot from audience perspectives and incorporates 5.5 hours of footage.

and 12 Pro devices, we will demonstrate artistic and technical details, including multi-perspective 2.5D video of actor performances, remote control integrated with theater systems, and our fiducial-based relocation with elements of the theater set. An example of a character looking through their *Grasshopper*, as seen by the audience (also looking through a device) is shown in Figure 2.

"IRL" elements. The demonstration will integrate physical props (beyond *Grasshoppers* in their custom enclosures), theater lighting, and sound from the show. Visitors and the authors will stand in for on-stage performers to convey the experience.

Operator view. We will show a live operator view that visualizes tracking of audience devices integrated with layers of behind-the-scenes data, including previsualization and fiducial alignment. (See Figure 3.) It illustrates our approach of using the virtual world for purposes beyond the direct audience experience.

Process documentation. Separate monitors will show additional video material illustrating details of the multidisciplinary design, testing, and production process.

Interactive film. When the production was required to move to a "live-to-tape" approach due to pandemic restrictions, we developed a full-length interactive streaming film using the recordings, built on the Cinema8 platform, to convey the nature of the live experience. It offers branching choices among storylines and offers instant switching between multiple audience camera views and audio feeds. (See Figure 4.) Using the interactive film, attendees will have the option to view any scene from the entire show, making their own "cut" by choosing storylines and perspectives.

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¹Created in our 2018 *Future Storytelling Summer Institute*; see vimeo.com/315600478